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Comparative Evaluation of Immersive Virtual Reality and Document-Based Training for Industrial Gas Leak Response

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Outline

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Introduction

Problem

- Natural gas processing involves high-pressure, safety-critical operations.
- Conventional training uses text-based SOPs and manuals.
- Text lacks contextual realism, spatial cues, and urgency.

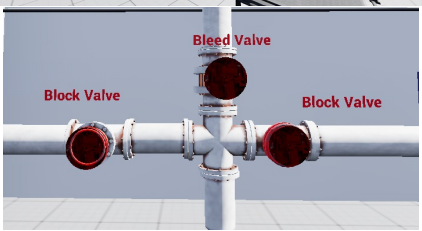
Proposed Solution

- Immersive VR-based guided learning for gas plant workers.
- Simulates an industrial Acid Gas Removal Unit (AGRU).
- Integrates DBB and LOTO protocols interactively.

Research Goal

- Compare VR-Based vs. Document-Based training.





Literature Review

VR in Safety-Critical Training

- Consistent gains in engagement, procedural recall, and decision-making (Stefan et al.,2023)
- Pre/post-test design effectively validates cognitive gains.

VR in High-Risk Industries

- AR/VR improves hazard recognition and reduces errors (Al-Khafaji et al.,2026)
- Text-based docs fail to replicate alarms and sensory overload (Mehta et al.,2025)

Research Gap

- No prior work unifies DBB protocol, LOTO procedures, and spatial AV fault localization in a single VR safety training framework.

Problem Formulation and Hypothesis

Problem Statement

- Responders must follow structured SOPs in sequence:
  1. SCBA deployment.
  2. Leak localization via AV cues.
  3. Double Block and Bleed (DBB).
  4. Lockout-Tagout (LOTO).
  5. Fault identification.
- Conventional training cannot replicate urgency or spatial awareness.

Hypotheses

- **H_1:** VR training yields significantly higher procedural recall and confidence than document-based training.
- **H_0:** No statistically significant difference between training outcomes.

Fault Scenarios Tested

- Stem leakage
- Gasket failure
- Pipe cracking

VR Training Environment

Hardware and Software

- **Engine:** Unreal Engine 5.4.4
- **HMD:** Meta Quest 3
- **GPU:** NVIDIA GeForce RTX A5000

Key Innovation

- Fault localization uses spatial audio hissing and visual gas emissions to guide workers to the exact leak before initiating DBB.



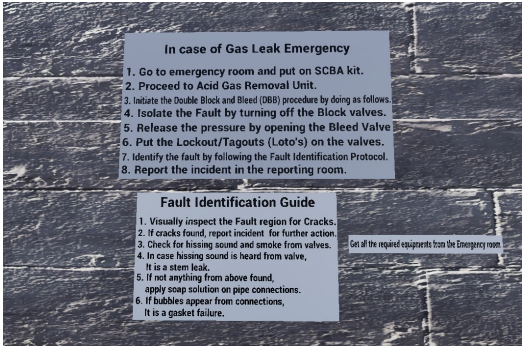




5-Stage VR Training Workflow

The 5 Stages

1. **Supervision Room:** SOP briefing and emergency protocol reference.
2. **Alarm Event:** Synchronized audio alarm and visual warning.
3. **SCBA Deployment:** Emergency equipment donning.
4. **Leak Localization:** Spatial audio and visual cues guide user.
5. **Isolation:** DBB , LOTO , Incident report.



Experimental Design

Participants

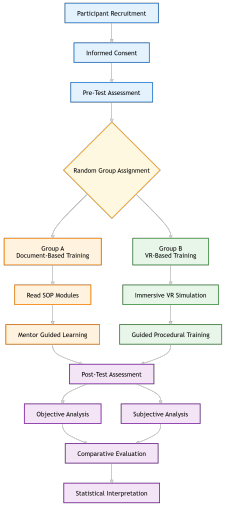
- $N = 30$ individuals, age 18–24 years.
- No prior industrial safety experience.
- **Group A** ($n = 15$): Document-Based.
- **Group B** ($n = 15$): VR-Based.

Evaluation Metrics

- **Knowledge:** $K = \frac{S}{15} \times 100$
- **Gain:** $G_k = S_{\text{post}} - S_{\text{pre}}$
- **Confidence:** $G_c = C_{\text{post}} - C_{\text{pre}}$

Assessment Topics

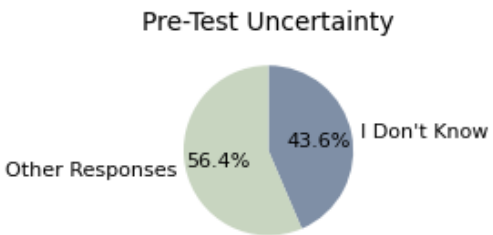
- SCBA deployment and hazard zone.
- Fault localization and identification.
- DBB protocol and LOTO application.
- Emergency decision-making.



Results – Pre-Test Baseline

High Uncertainty at Baseline

- The "I Don't Know" response was selected **43.6%** of the time across all participants.
- Confirms very limited baseline knowledge of gas emergency protocols.
- Both groups were statistically comparable at baseline (no prior advantage).



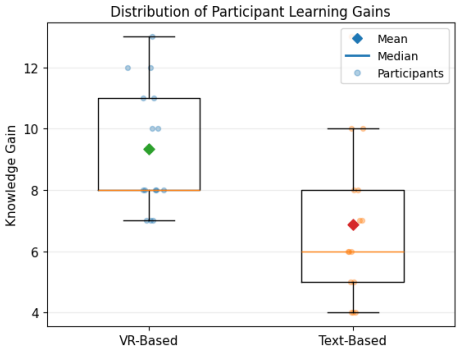
Results – Knowledge Accuracy and Learning Gain

Post-Training Performance

Metric	Doc-Based	VR-Based	Difference
Pre-Test Score	3.53	3.13	–
Post-Test Score	10.40	**12.47**	+2.07
Learning Gain	6.87 pts	**9.34 pts**	+2.47
Post Accuracy (%)	69.33%	**83.11%**	+13.78%

Statistical Analysis

- Welch's t-test: $p < 0.01$ (Significant).
- VR training achieved **+13.78%** higher accuracy.



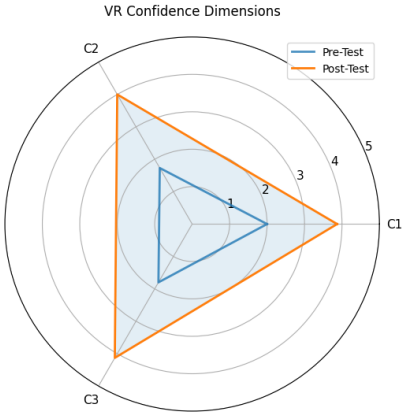
Results – Confidence and Section-Wise Analysis

Confidence Improvement

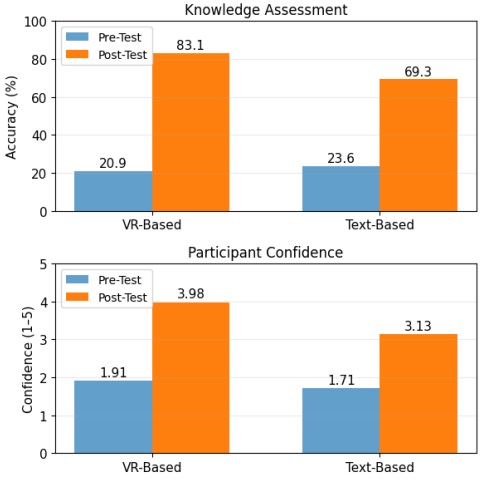
Metric	Doc	VR
Pre-Training	1.71	1.91
Post-Training	3.13	**3.98**
Gain	1.42	**2.07**

Observations

- VR outperforms in all sections.
- Largest gain in DBB Protocol.
- Confidence gain is 45% higher in VR group.



Results – Overall Comparison



Discussion

Why VR Outperforms Document-Based Training

- **Embodied Learning:** Workers physically enact SOPs rather than passively reading.
- **Multimodal Cues:** Spatial audio and gas visuals create a richer learning context.
- **Real-time Feedback:** Immediate consequences reinforce correct decision chains.
- **Urgency Simulation:** Alarms replicate actual emergency stressors.

Statistical Validation

- Welch's two-sample t-test on learning gains: $p < 0.01$.
- H_0 rejected; H_1 supported.
- VR group showed lower inter-subject variance.

Limitations

- Scale: Small controlled sample ($N = 30$) of academic students.
- Retention: Only immediate post-training outcomes assessed.
- Platform: Meta Quest 3 hardware required.

Conclusion

Key Contributions

- Developed an immersive VR safety training framework simulating a real Acid Gas Removal Unit (AGRU) facility.
- Uniquely integrated DBB and LOTO protocols with multimodal spatial AV fault localization in a VR workflow.
- Conducted a rigorous controlled comparative study with objective scoring, Likert confidence analysis, and Welch's t-test.

Core Results at a Glance

Metric	Doc-Based	VR-Based	Significance
Post Accuracy	69.33%	**83.11%**	$p < 0.01$
Learning Gain	6.87 pts	**9.34 pts**	
Confidence Gain	1.42	**2.07**	

Outcome: VR training significantly outperforms document-based instruction for safety-critical tasks.

Thank You!

Any Questions?

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github.com/Praso28/ICVARS_D046